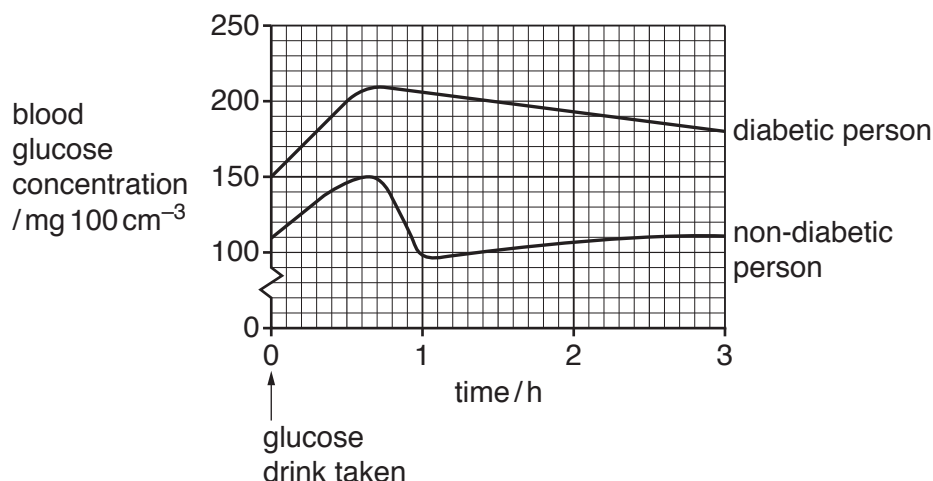


- 7 (a) Some people have a condition called diabetes. In type 1 diabetes the pancreas does not produce enough insulin.

Fig. 7.1 shows the blood glucose concentrations of a type 1 diabetic person and a non-diabetic person, at regular intervals after drinking a glucose drink.



**Fig. 7.1**

- (i) Describe the results shown in Fig. 7.1.

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..... [3]

- (ii) Name the location of the receptors in a non-diabetic person that detect a change in blood glucose concentration.

..... [1]

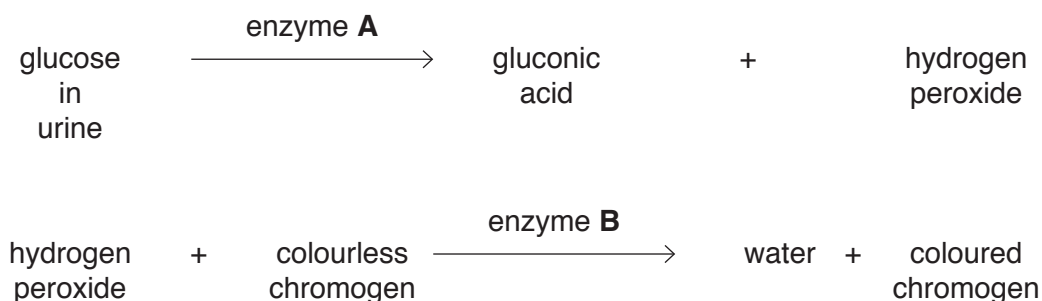
- (iii) Name the homeostatic mechanism by which blood glucose concentration is maintained.

..... [1]

- (b) The urine of a non-diabetic person does not contain glucose. A person with type 1 diabetes will excrete glucose in urine.

A reading of the concentration of glucose in the urine can be estimated using a dipstick.

Fig. 7.2 outlines how a dipstick works.



**Fig. 7.2**

The higher the concentration of glucose in the urine, the darker the colour on the dipstick.

- (i) Name enzymes **A** and **B**.

**A** .....

**B** ..... [2]

- (ii) An electronic biosensor can be used to measure the glucose concentration in a drop of blood.

Suggest **one** advantage of using a biosensor and **one** advantage of using a dipstick to measure glucose concentration.

*biosensor* .....

.....

*dipstick* .....

..... [2]

(c) Describe the role played by insulin in the control of blood glucose concentration.

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..... [5]

[Total: 14]